

EEE554 – Probability and Random Processes

Module 7: Functions of Random Variables

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From Random Variables to Functions of Random Variables

In the previous modules, we introduced a random variable as a map

$$X : \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto X(\omega).$$

In engineering, we are often interested not directly in X , but in a quantity obtained by processing X .

For example,

$$Y = g(X).$$

Examples:

- Received power $P = V^2$
- Signal-to-noise ratio $\gamma = \frac{P}{N}$
- Doppler shift $f_D = f_c \frac{v}{c} \cos \theta$
- Signal energy $E = \sum_i X_i^2$

Question:

If we know the distribution of X , what is the distribution of $Y = g(X)$?

Functions of Random Variables

Suppose X is a random variable and

$$Y = g(X),$$

where

$$g : \mathbb{R} \rightarrow \mathbb{R}.$$

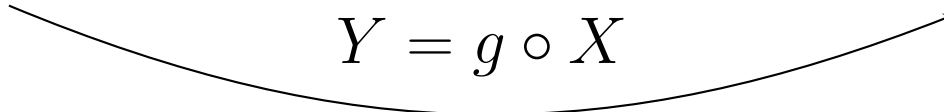
Then Y is also a random variable.

Indeed, by definition,

$$X = X(\omega).$$

Therefore,

$$Y(\omega) = g(X(\omega)) = (g \circ X)(\omega).$$

$$\omega \in \Omega \xrightarrow{X} X(\omega) \in \mathbb{R} \xrightarrow{g} Y(\omega) = g(X(\omega)) \in \mathbb{R}$$

$$Y = g \circ X$$

Hence,

$$Y : \Omega \rightarrow \mathbb{R}$$

is also a random variable.

Examples

$$Y = X^2,$$

$$Y = \sqrt{X},$$

$$Y = \log X,$$

$$Y = \cos X.$$

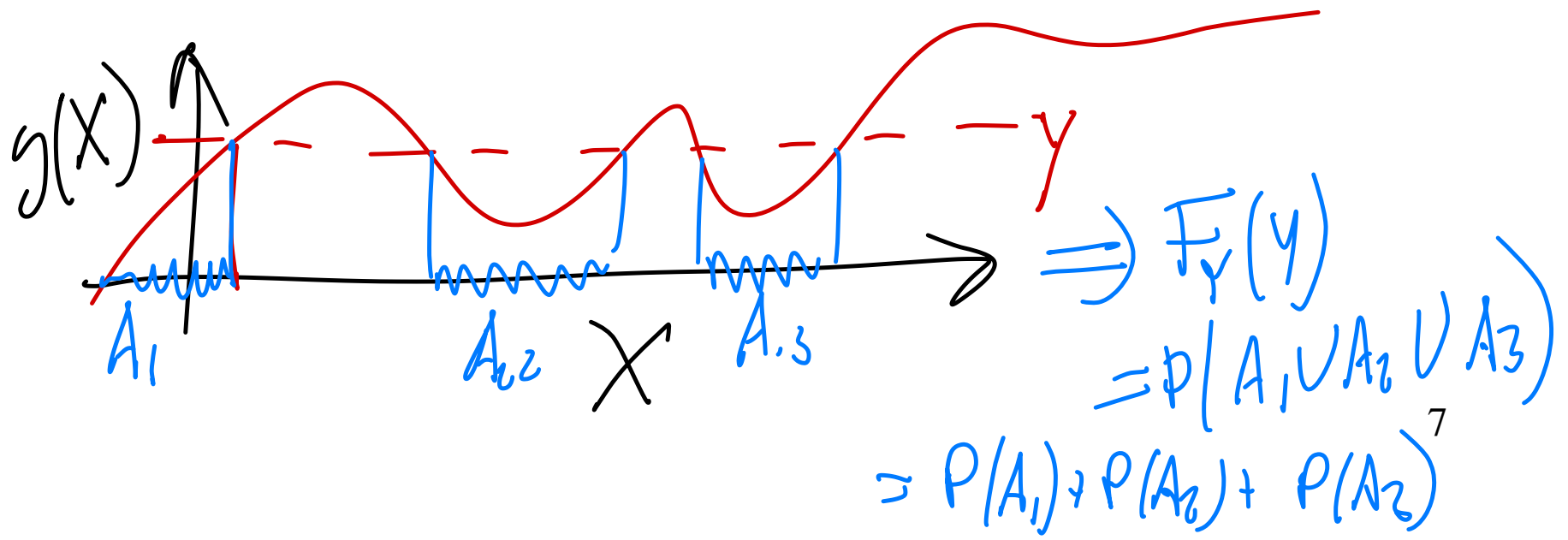
CDF Method

The distribution of Y is obtained from

$$F_Y(y) = P(Y \leq y).$$

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leq y).$$



The key step is to rewrite

$$\{g(X) \leq y\}$$

as an event involving X :

$$A_y: \{X: g(X) \leq y\}$$

$$\boxed{\{g(X) \leq y\} = \{X \in A_y\}.}$$

Then

$$F_Y(y) = P(X \in A_y).$$

Once $F_Y(y)$ is known, differentiate (when possible) to obtain

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

General Procedure

To compute the distribution of $Y = g(X)$,

1) Sketch (if helpful):

$$y = g(x).$$

2) For a fixed y , rewrite the event

$$\{g(X) \leq y\}$$

as

$$\{X \in A_y\}$$

for some set A_y depending on y .

3) Then compute

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \in A_y).$$

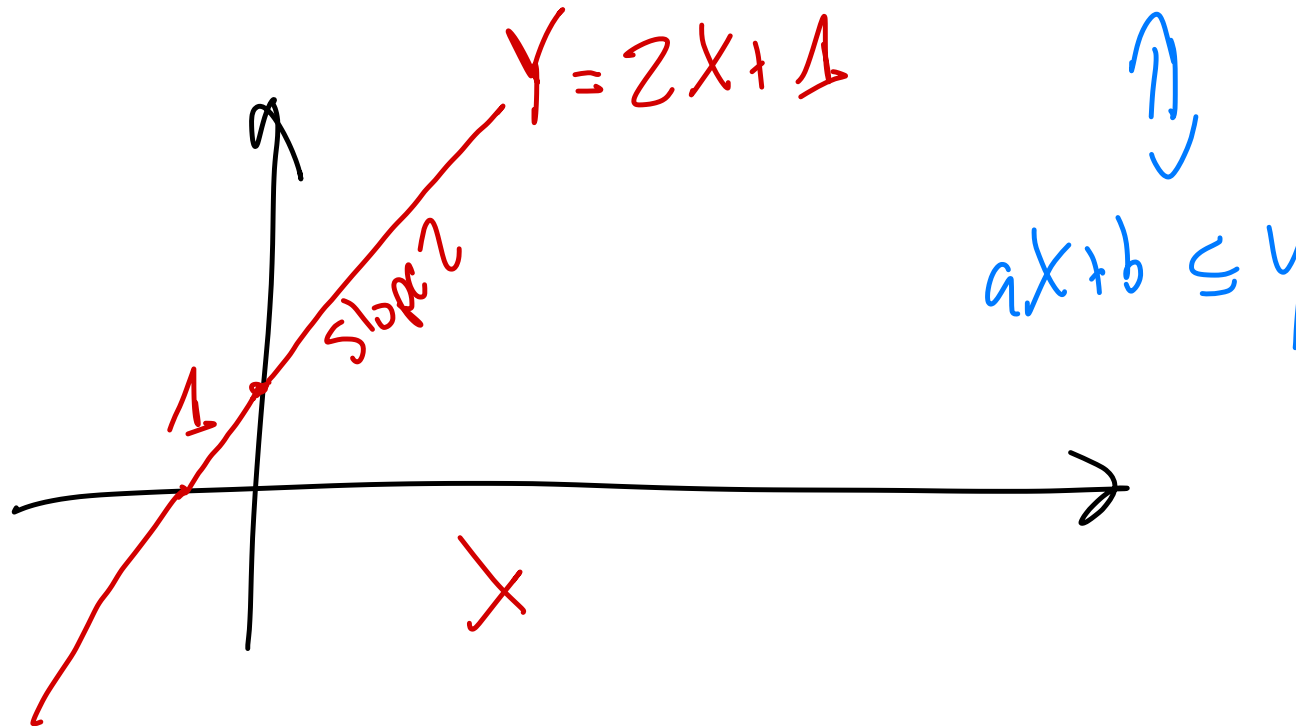
4) Differentiate $F_Y(y)$ with respect to y to obtain the PDF $f_Y(y)$, when it exists.

Example 1: Affine Transformation

Suppose X is a continuous random variable and

$$Y = aX + b, \quad a \neq 0.$$

Find $f_Y(y)$.



$$aX + b \leq y \Leftrightarrow X \leq \frac{y-b}{a}$$

Solution: If $a > 0$,

$$F_Y(y) = P\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

If $a < 0$,

$$F_Y(y) = 1 - F_X\left(\frac{y-b}{a}\right).$$

Differentiating,

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

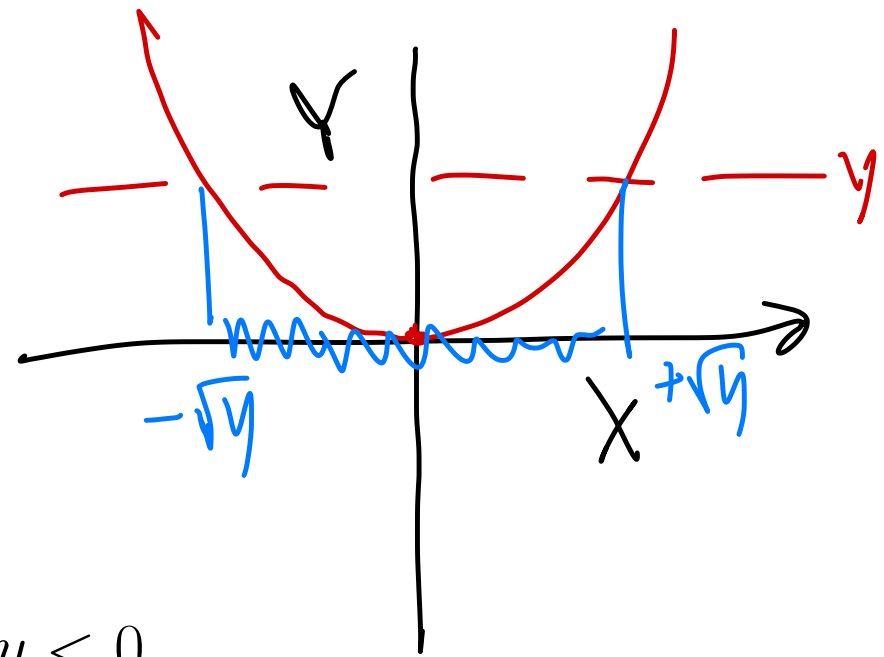
$$aX + b \leq y \Leftrightarrow X \geq \frac{y-b}{a}$$

$$\begin{aligned} & \stackrel{!}{=} P\left(X \geq \frac{y-b}{a}\right) = 1 - P\left(X < \frac{y-b}{a}\right) \\ & \stackrel{!}{=} 1 - F_X\left(\frac{y-b}{a}\right) \end{aligned}$$

Example 2

Suppose

$$Y = X^2.$$



Notice that $Y \geq 0$, hence $F_Y(y) = 0$ for $y < 0$.

If $y \geq 0$,

$$F_Y(y) = \overset{= P(Y \leq y)}{P(X^2 \leq y)}$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y})$$

$$= F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

$$= P(X \leq \sqrt{y}) - P(X < -\sqrt{y})$$

Differentiating,

$$f_Y(y) = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}, \quad y > 0.$$

Interpretation.

The transformation

$$Y = X^2$$

is not one-to-one.

For every $y > 0$, two possible values of X ,

$$X = \sqrt{y} \quad \text{and} \quad X = -\sqrt{y},$$

produce the same value

$$Y = y.$$

Therefore, both values contribute to the PDF of Y .

Monotonic Transformation

Suppose $Y = g(X)$, where g is one-to-one and differentiable.

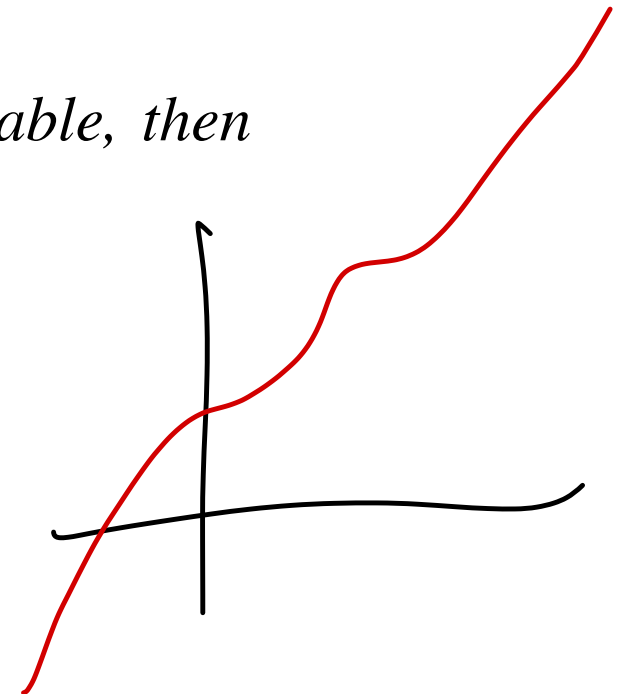
Assume first that g is strictly increasing.

Theorem 1. *If g is strictly increasing and differentiable, then*

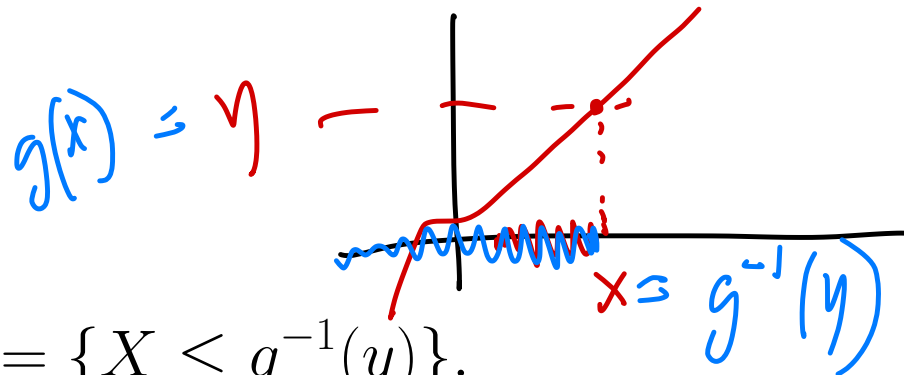
$$F_Y(y) = F_X(g^{-1}(y)).$$

If, in addition, g^{-1} is differentiable, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$



Proof. Since g is strictly increasing,



$$\{Y \leq y\} = \{g(X) \leq y\} = \{X \leq g^{-1}(y)\}.$$

Therefore,

$$F_Y(y) = P(Y \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

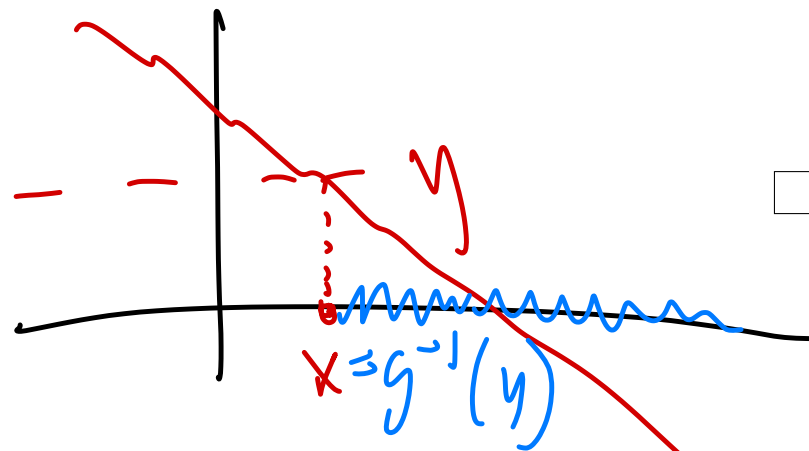
Differentiating with respect to y ,

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

Since g^{-1} is increasing, the derivative is positive, so this equals

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

If g is strictly decreasing, then



$$\{Y \leq y\} = \{g(X) \leq y\} = \{X \geq g^{-1}(y)\}.$$

Hence,

$$F_Y(y) = 1 - F_X(g^{-1}(y)).$$

Differentiating gives the same PDF formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

$$\left(\frac{dg(x)}{dx} \right)^{-1} \Big|_{x=g^{-1}(y)}$$

Why Does the Derivative Appear?

A transformation

$$y = g(x)$$

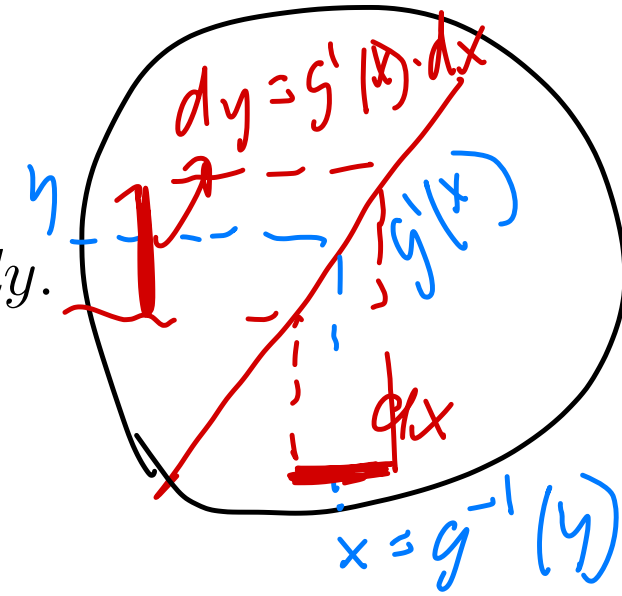
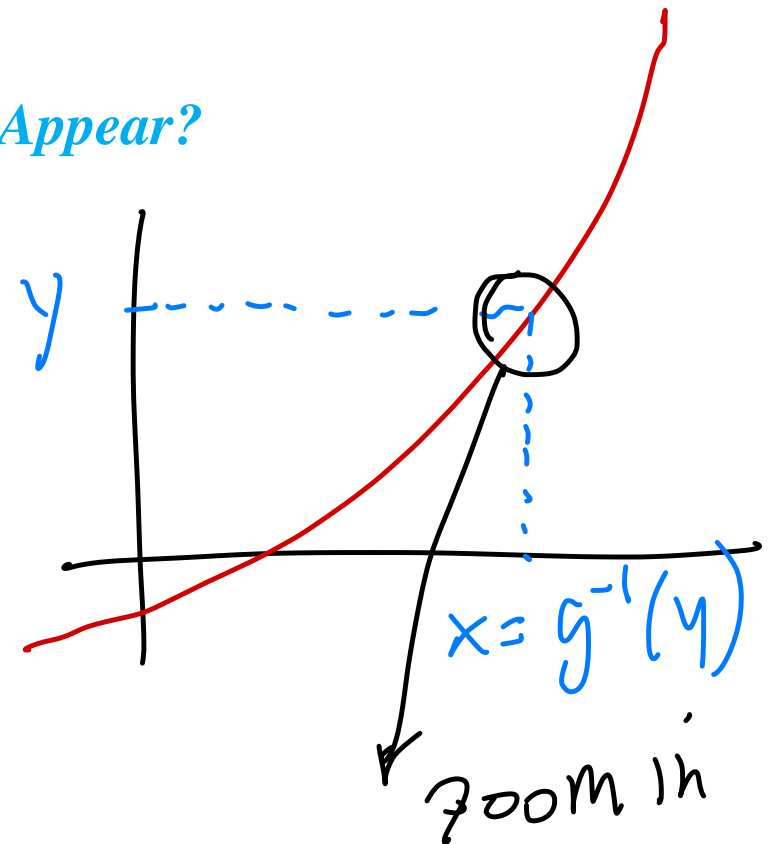
locally stretches or compresses intervals.

For a small interval dx ,

$$dy \approx |g'(x)| dx.$$

Probability must be preserved:

$$f_X(x) dx \approx f_Y(y) dy.$$



Therefore,

$$f_Y(y) \approx f_X(x) \left| \frac{dx}{dy} \right|.$$

Hence,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

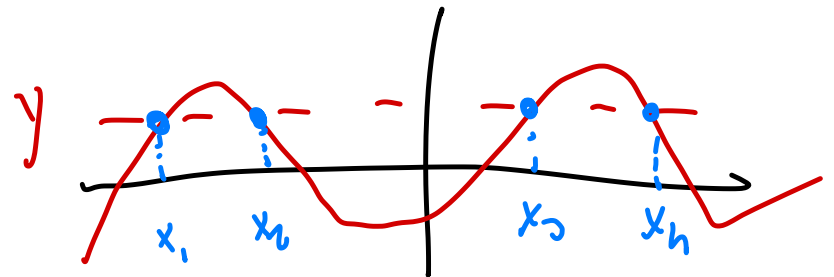
The derivative accounts for the local stretching or compression introduced by the transformation.

Non-Monotonic Transformations

The previous result can be generalized when g is not one-to-one.

Suppose the equation

$$g(x) = y$$



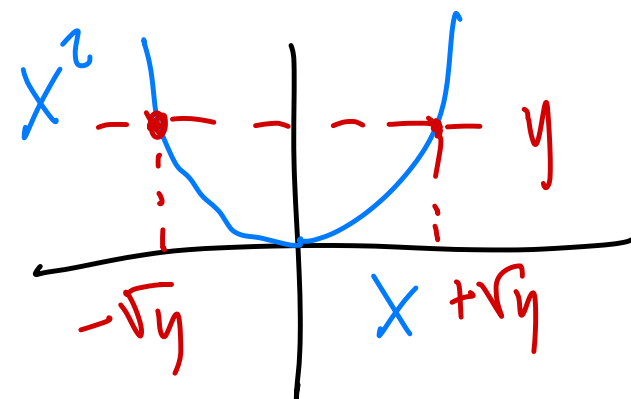
has multiple solutions $x_1, x_2, \dots$. Then each solution contributes to the probability density at y :

$$f_Y(y) = \sum_i f_X(x_i) \left| \frac{dx_i}{dy} \right|,$$

called the Jacobian formula.

The example

$$Y = X^2$$



is a special case, with

$$x_1 = \sqrt{y}, \quad x_2 = -\sqrt{y}.$$

$$x = g^{-1}(y) = \pm \sqrt{y}$$

$$x = \sqrt{y} \Rightarrow \left| \frac{dx}{dy} \right| = \frac{1}{2\sqrt{y}}$$

$$x = -\sqrt{y} \Rightarrow \left| \frac{dx}{dy} \right| = \frac{1}{2\sqrt{y}}$$

$$f_Y(y) = \frac{f_X(-\sqrt{y})}{2\sqrt{y}} + \frac{f_X(+\sqrt{y})}{2\sqrt{y}}$$

Engineering Example: Doppler Shift

A vehicle moves at constant speed a .

The angle between the direction of motion and the line of sight to the receiver is

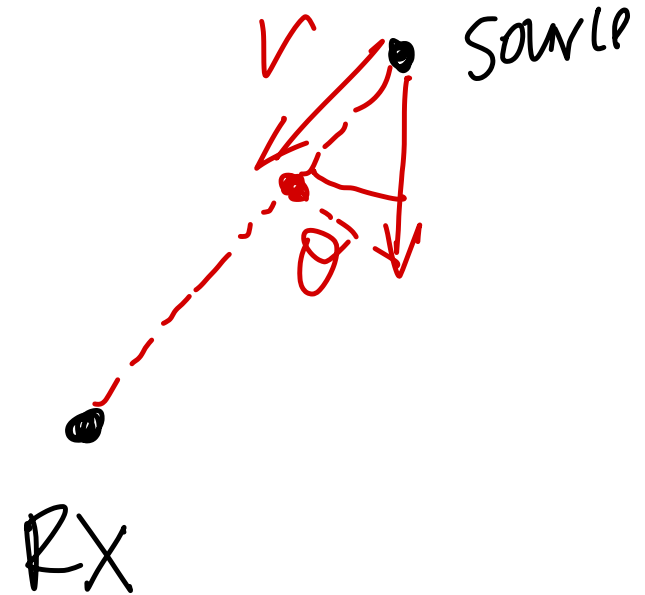
$$\Theta \sim \mathcal{U}[0, \pi].$$

The radial velocity is

$$V = a \cos \Theta.$$

The corresponding Doppler shift is

$$D = \frac{f}{c} V = D_{\max} \cos \Theta,$$



where

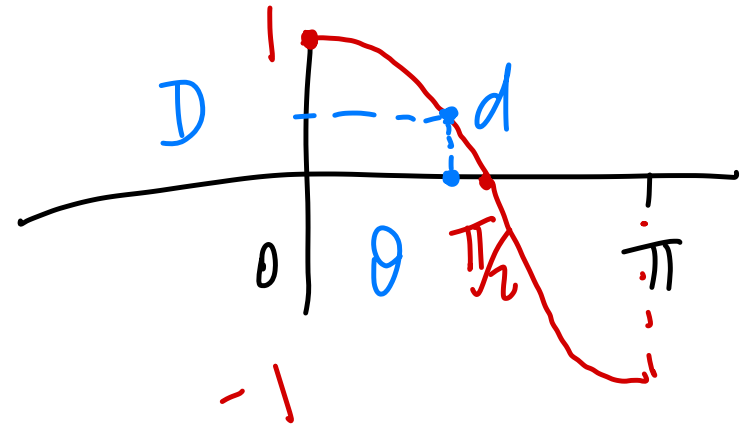
- f_c is the carrier frequency,
- c is the speed of light,
- $D_{\max} = \frac{f}{c}a$ is the maximum Doppler shift.

Since

$$D = D_{\max} \cos \Theta, \Rightarrow d = D_{\max} \cdot \cos \theta = g(\theta)$$

and $\cos \theta$ is one-to-one on $[0, \pi]$,

$$\theta = \arccos \left(\frac{d}{D_{\max}} \right) \cdot \frac{dg(\theta)}{d\theta} \Big|_{\theta = g^{-1}(d)} = -D_{\max} \cdot \sin \left(\arccos \left(\frac{d}{D_{\max}} \right) \right)$$



$$\left| \frac{dg(\theta)}{d\theta} \right|_{\theta=g^{-1}(d)} = D_{\max} \left| \sin \left(\arccos \left(\frac{d}{D_{\max}} \right) \right) \right|$$

$$|\sin| = \sqrt{1 - \cos^2}$$

$$= D_{\max} \cdot \sqrt{1 - \cos^2 \left(\arccos \left(\frac{d}{D_{\max}} \right) \right)}$$

$$= D_{\max} \cdot \sqrt{1 - \left(\frac{d}{D_{\max}} \right)^2} = \sqrt{D_{\max}^2 - d^2}$$

$$f_D(d) = f_\theta(g^{-1}(d)) \cdot \left| \frac{dg(\theta)}{d\theta} \right|_{\theta=g^{-1}(d)}$$

$$= \frac{1}{\pi} \cdot \frac{1}{\sqrt{D_{\max}^2 - d^2}}$$

Therefore,

$$\left| \frac{d\theta}{dd} \right| = \frac{1}{\sqrt{D_{\max}^2 - d^2}}.$$

Since

$$f_{\Theta}(\theta) = \frac{1}{\pi}, \quad 0 \leq \theta \leq \pi,$$

we obtain

$$f_D(d) = \frac{1}{\pi \sqrt{D_{\max}^2 - d^2}}, \quad |d| < D_{\max}.$$

Interpretation.

The randomness in the angle of motion induces randomness in the observed Doppler shift.

A deterministic physical relationship

$$D = D_{\max} \cos \Theta$$

transforms the distribution of Θ into the distribution of D .

Takeaway

To determine the distribution of

$$Y = g(X),$$

- use the CDF method,
- differentiate to obtain the PDF,
- use the Jacobian formula whenever applicable.

Many engineering problems reduce to transforming random variables.

